

1)

$$\sum x = 202 + 120 + 201 + 322 + 167 + 273 + 406 + 345$$

$$= 2056$$

$$\bar{x} = \frac{\sum x}{n} = \frac{2056}{8} = \underline{\underline{257}}$$

2)

$$\bar{x} = \frac{\sum x}{n}$$

$$13 = \frac{7+9+11+13+x+21}{6}$$

$$13 = \frac{61+x}{6}$$

$$13 \times 6 = \cancel{6} \times \underline{61+x}$$

$$78 = 61+x$$

$$x = 78 - 61$$

$$x = \underline{\underline{17}}$$

3)

$$\bar{x} = 35 \text{ kg}$$

$$N = 8$$

$$35 = \frac{\text{Total hlt}}{8}$$

$$35 \times 8 = \text{total hlt} = 280$$

$$\begin{aligned} \text{Weight of 7 girls} &= \cancel{63} + \cancel{106} + 35 + 28 + 43 + 25 + 33 + 51 + 22 \\ &= 237 \end{aligned}$$

$$\begin{aligned} \text{Weight of 8}^{\text{th}} \text{ girl} &= \text{Total weight} - \text{Weight of 7} \\ &= 280 - 237 \\ &= \underline{\underline{43 \text{ kg}}} \end{aligned}$$

4)

$$\begin{aligned} \bar{x} = m + 2 &= \frac{(2m + 3) + (m + 2) + (3m + 4) + (4m + 5)}{4} \\ &= (2 + 1 + 3 + 4)m = 10m - 6 \end{aligned}$$

$$3 + 2 + 4 + 5 = 14$$

$$m + 2 = \frac{10m + 14}{4}$$

$$4(m+2) = 10m + 14$$

$$4m + 8 = 10m + 14$$

$$8 = 6m + 14$$

$$-6 = 6m$$

$$m = -1$$

5)

$$1445 = \frac{\text{Total}}{10}$$

$$\text{total Sal of 10 workers} = 1445 \times 10$$

$$= \underline{\underline{14450}}$$

$$11 \text{ worker} = \underline{\underline{15950}}$$

$$\text{new mean} = \frac{15950}{11}$$

$$= \underline{\underline{1450}}$$

6) x f dx

5 4 20

10 5 50

15 7 105

20 4 80

25 3 75

30 2 60

$$390 = \sum dx$$

$$\sum f = 25$$

$$\bar{x} = \frac{\sum dx}{\sum f} = \frac{390}{25} = 15.6$$

7) x f dx

14 2 28

36 5 180

45 1 45

70 3 210

$$\frac{11}{463} = \sum dx$$

$$11 = \sum f$$

$$\bar{x} = \frac{\sum dx}{\sum f} = \frac{463}{11}$$

$$= \underline{\underline{42.09}}$$

8)

$$\text{mean} = \frac{850 + 700 + 100 + 750 + 5000 + 180 + 420 + 2500 + 400 + 360}{10}$$

$$= \frac{11160}{10} = \underline{\underline{1116}} = \text{Mean Family Income}$$

9)

marks(x)	d	dx	
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64	8	512	
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63	18	1134	
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62	12	744	
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61	9	549	
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60	7	420	
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59	6	354	
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$$\Sigma d = 60$$

$$\Sigma dx = 3713$$

$$\bar{X} = \frac{\Sigma dx}{\Sigma d} = \frac{3713}{60} = \underline{\underline{61.88}}$$

Median

$$= \frac{N}{2} = \frac{60}{2} = 30$$

marks	f	Cf
59	6	6
60	7	13
61	9	22
62	12	34
63	18	52
64	8	60

median = 62

10)

x	f	fx
100	5	500
140	2	280
170	6	1020
200	4	800
250	3	750
	<u>20</u>	<u>3350</u>

$$\sum f = 20$$

$$\sum fx = 3350$$

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{3350}{20}$$

$$= \underline{\underline{167.5}}$$